

Sequences & Series

ANSWER KEY



Arithmetic sequences

- 1 For the arithmetic sequence with $a_1 = 6$ and common difference $d = 4$, find a_{12} .

$$a_{12} = a_1 + (12 - 1)d = 6 + 11(4) = 50.$$

- 2 For the arithmetic sequence with $a_1 = 20$ and $d = -3$, find a_{15} .

$$a_{15} = 20 + 14(-3) = 20 - 42 = -22.$$

- 3 A geometric sequence has $a_1 = 3$ and each term is 2 times the previous one. Find the smallest n for which $a_n = 384$.

$$a_n = 3 \cdot 2^{n-1}. \text{ Setting } 3 \cdot 2^{n-1} = 384 \text{ gives } 2^{n-1} = 128 = 2^7, \text{ so } n - 1 = 7 \text{ and } n = 8.$$

Arithmetic series (sum formula)

- 4 Find S_{12} for the arithmetic sequence with $a_1 = 6$, $d = 4$ (using $a_{12} = 50$ from above).

$$S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(6 + 50)}{2} = 6(56) = 336.$$

- 5 Find S_8 for the arithmetic sequence with $a_1 = 2$, $d = 5$.

$$a_8 = 2 + 7(5) = 37. S_8 = \frac{8(2 + 37)}{2} = 4(39) = 156.$$

Geometric sequences

- 6 For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_6 .

$$a_6 = a_1 r^5 = 4(3^5) = 4(243) = 972.$$

- 7 A geometric sequence begins 5, -10 , 20 , -40 , Find r and a_7 .

$$r = \frac{-10}{5} = -2. a_7 = 5(-2)^6 = 5(64) = 320.$$

Geometric series (finite)

- 8 Find S_6 for the geometric sequence with $a_1 = 4$, $r = 3$ (using $a_6 = 972$ from above).

$$S_6 = \frac{a_1(r^6 - 1)}{r - 1} = \frac{4(729 - 1)}{2} = \frac{4(728)}{2} = 1456.$$

- 9 Find S_5 for the geometric sequence $5, -10, 20, -40, \dots$

$$S_5 = \frac{5[(-2)^5 - 1]}{-2 - 1} = \frac{5(-32 - 1)}{-3} = \frac{5(-33)}{-3} = 55.$$

Infinite geometric series

- 10 Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \dots$.

$$a = 20, r = \frac{1}{3}, \text{ and } |r| < 1, \text{ so by Theorem 3.2 the sum is } \frac{20}{1 - \frac{1}{3}} = \frac{20}{\frac{2}{3}} = 30.$$

- 11 Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \dots$.

$$a = 9, r = -\frac{1}{3}, \text{ and } |r| < 1, \text{ so the sum is } \frac{9}{1 - (-\frac{1}{3})} = \frac{9}{\frac{4}{3}} = \frac{27}{4}.$$

- 12 Explain why the infinite geometric series with $a = 5, r = 2$ has no finite sum.

Theorem 3.2 requires $|r| < 1$ for the partial sums to converge. Here $|r| = 2 > 1$, so the terms grow without bound and the series diverges — it has no finite sum.